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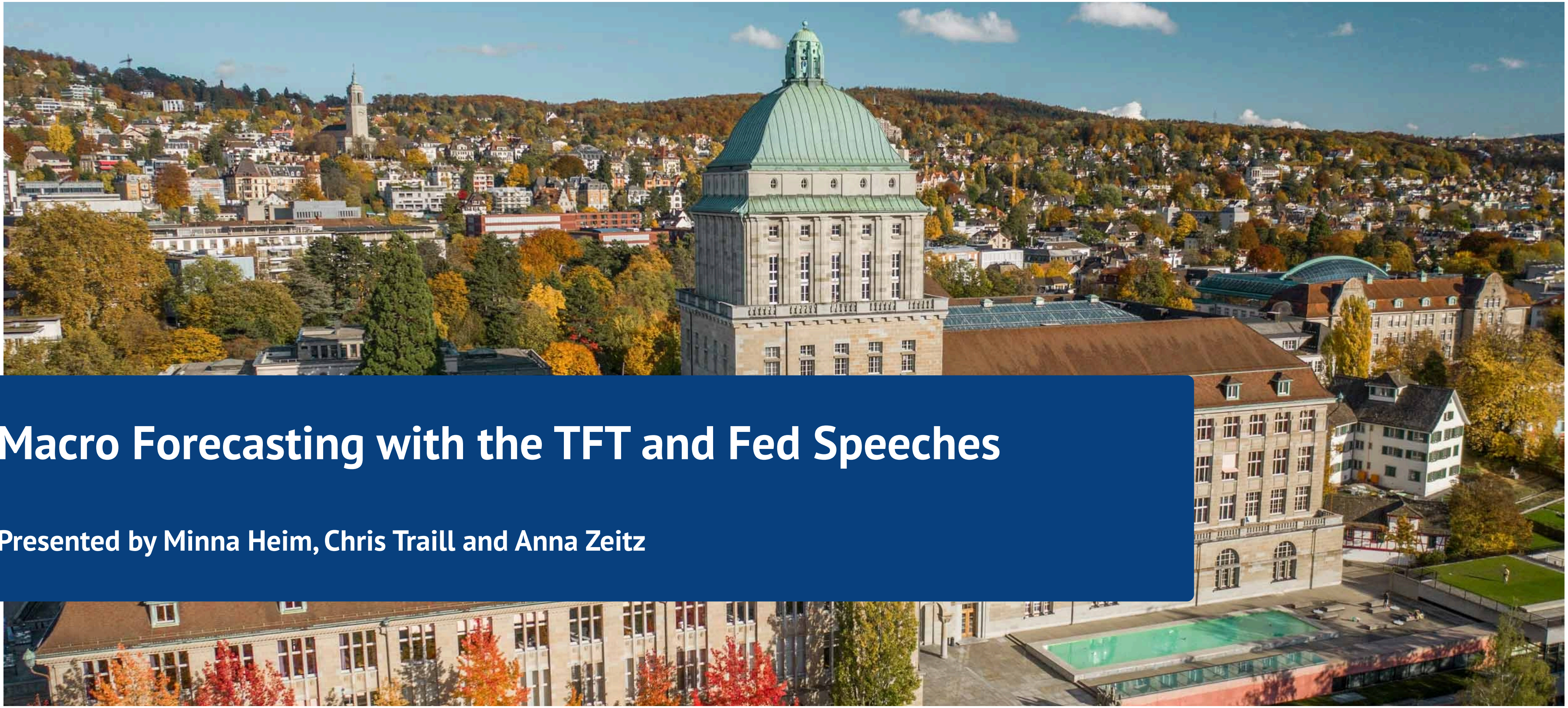

Macro Forecasting with the TFT and Fed Speeches

Presented by Minna Heim, Chris Traill and Anna Zeitz

Motivation

- Macro forecasts matter for
  - Governments
  - Financial markets
  - **Central banks**

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$\Rightarrow$ Can we do better than standard macro models?

Research Question

- The Fed communicates with the public through official statements and speeches
- It may communicate additional information:
  1. **Superior information**: private data and internal models not available to the public
  2. **Forward guidance**: signals about future policy to guide market expectations

Research Question

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- It may communicate additional information:
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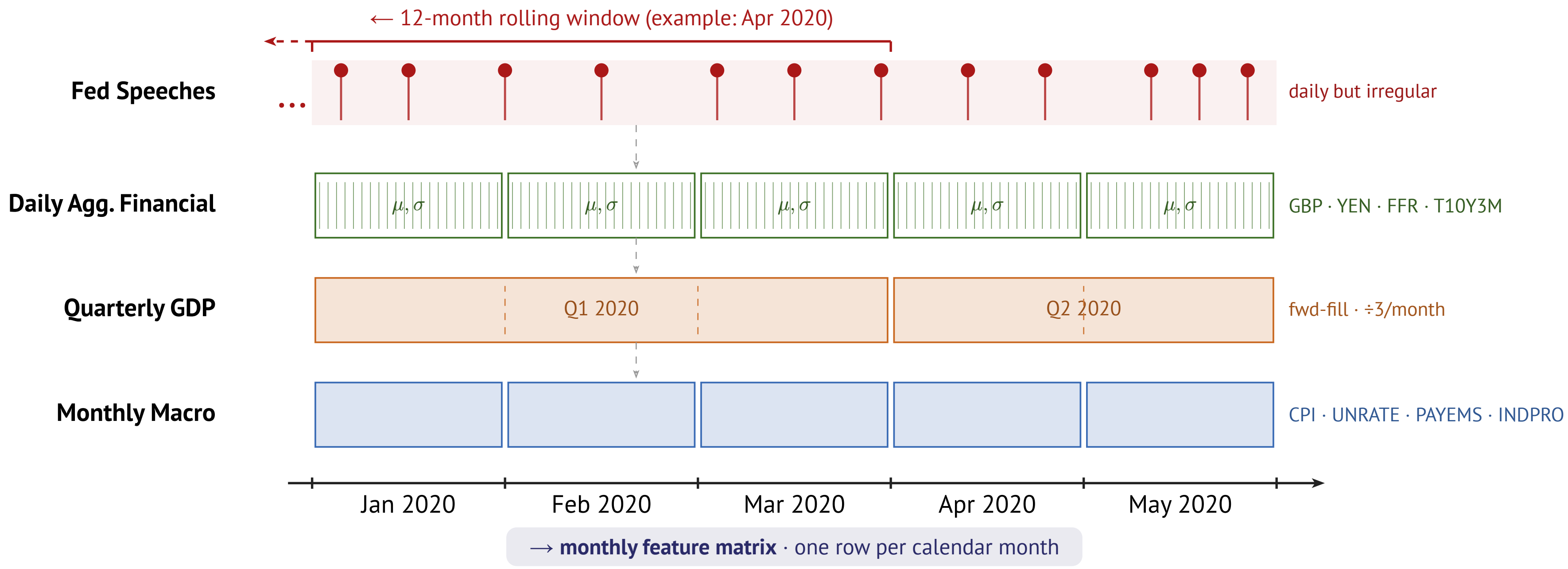
$\Rightarrow$ **Do Fed speeches contain information useful to forecast macroeconomic indicators?**

Data Alignment



Models

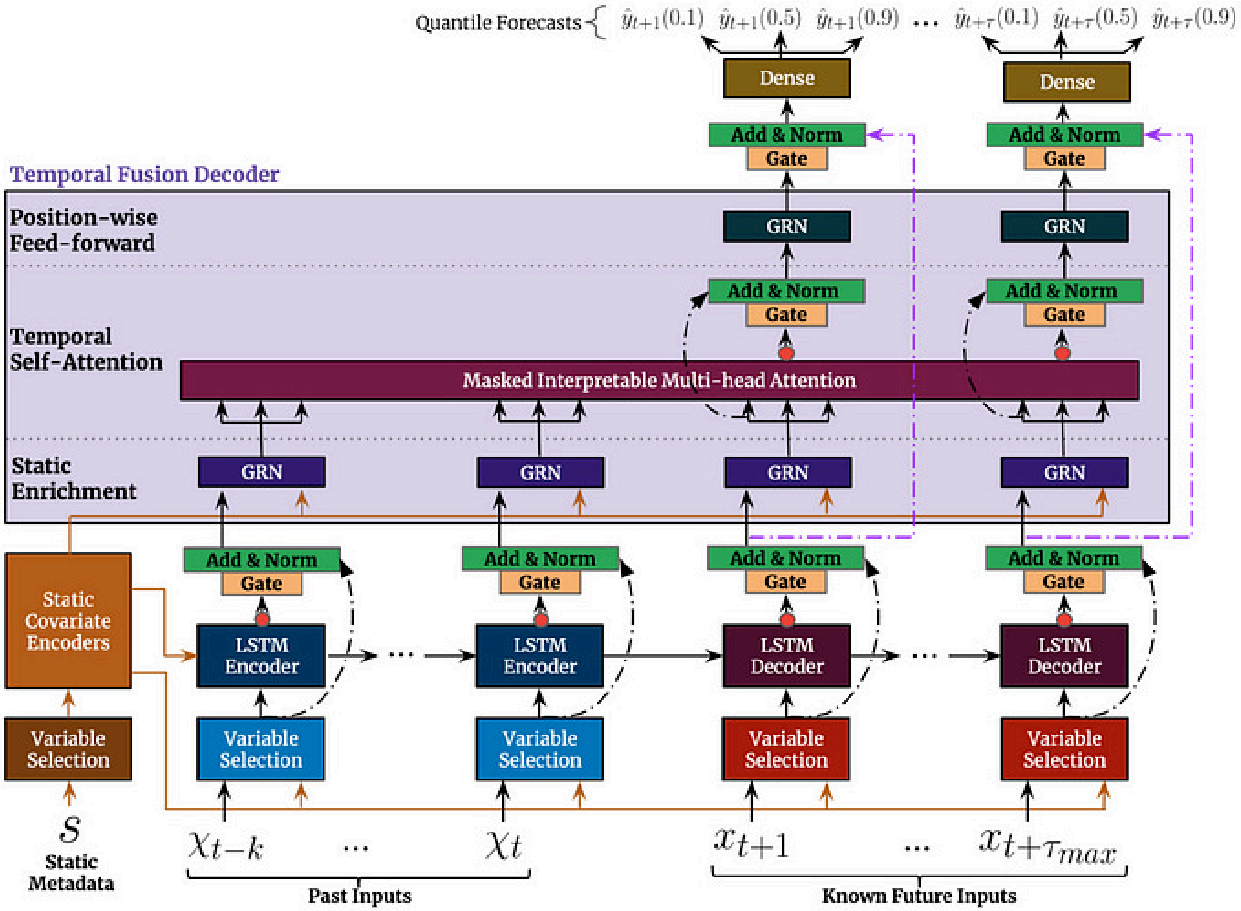
Benchmarks

$$\text{AR(p)} \quad y_t = c + \sum_{i=1}^{p} \varphi_i y_{t-i} + \varepsilon_t$$

$$\text{ARIMA(p, d, q)} \quad \Delta^d y_t = c + \sum_{i=1}^{p} \varphi_i \Delta^d y_{t-i} + \sum_{j=1}^{q} \theta_j \varepsilon_{t-j} + \varepsilon_t$$

- Integration: d-th difference of y_t
- MA component: q lags of residuals ε_t

Temporal Fusion Transformer



Speech Embeddings

Speech Embeddings

Two models chosen for complementary perspectives:

1. **FinBERT**: BERT-based, pre-trained on financial statements (Yang et al., 2020)
2. **FOMC-RoBERTa**: fine-tuned on FOMC data, hawkish/dovish classification (Shah et al., 2023)

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Processing: chunk-and-average (512-token windows, 128-token overlap)

- Assumption: information spread across entire speech, not just the beginning

Dimensionality reduction: 768 $\Rightarrow X$ dims via Principal Component Analysis or Factor Analysis

- X is treated as hyperparameter

Speech Alignment

Speeches are held at **irregular** and **daily** frequency: how to align with monthly horizon?

1. **Mean aggregation**: unweighted mean over rolling window of X months
2. **Exponential decay**: $w_t = \exp(-\lambda \cdot d_t)$ where d_t = days before month m
   - More recent speeches receive higher weight
   - λ treated as fixed; speech window X tuned as hyperparameter
3. **Attention-based**: key = embedding (after dimensionality reduction), query = mean macro state
   - Given current macro conditions, which past speeches are most informative?
   - Fitted on training data only $\Rightarrow$ no leakage!

All methods also track: voter speech ratio, chair speech ratio, gender share

Hyperparameter Tuning

Tune in two stages:

1. **Stage 1: Architecture (Macro-Only TFT)**
   - Bayesian search via Optuna (TPE sampler)
   - 2-fold CV, up to 50 trials
   - Includes: encoder length, hidden size, normalizer
2. **Stage 2: Speech Embedding Params**
   - Architecture fixed from Stage 1
   - Tune: aggregation, reduction, PCA dims, speech window

Evaluation Protocol

- **Walk-forward cross-validation**: expanding window, never use future data
- **Metric**: MAE (mean absolute error), RMSE ????

ADJUST HERE

- **Multi-step forecasting**: predictions in non-overlapping 12-month windows
  - Context: training data + model's own previous predictions
  - Same assumption for AR, ARIMA and TFT $\Rightarrow$ fair comparison

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$\Rightarrow$ All models evaluated on the same information set

Results

Results

- Incl. Tables, Figures

Robustness Checks

- Shuffling Embeddings
- Using unrelated text (German Texts of Franz Kafka)
- Use not full speech but only first 512 tokens of speech???

Future Work

- Hyperparameter tuning: switch to one stage tuning! Also: context-dependent attention
- Try out SSMs for forecasting
  - Combine with TFT: replace LSTM with SSM for sequence encoder
- Add Vintages: use real-time macro data (as available at forecast time) for more realistic evaluation
- Extend dataset: Working Paper (Lustenberger et al., 2026)
  - 10,000+ speeches since 1914, continuously updated

Bibliography

Lustenberger, T., Rossi, E., & Zeitz, A. (2026). *Central Bank Communication: New Data and Stylized Facts From a Century of Fed Speeches* (Working Paper No. 6/2026). https://doi.org/tbd

Shah, A., Paturi, S., & Chava, S. (2023,). Trillion Dollar Words: A New Financial Dataset, Task & Market Analysis. *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL)*. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4447632

Yang, Y., UY, M. C. S., & Huang, A. (2020,). *FinBERT: A Pretrained Language Model for Financial Communications*. https://arxiv.org/abs/2006.08097

Appendix

Variable List: Macroeconomic Variables

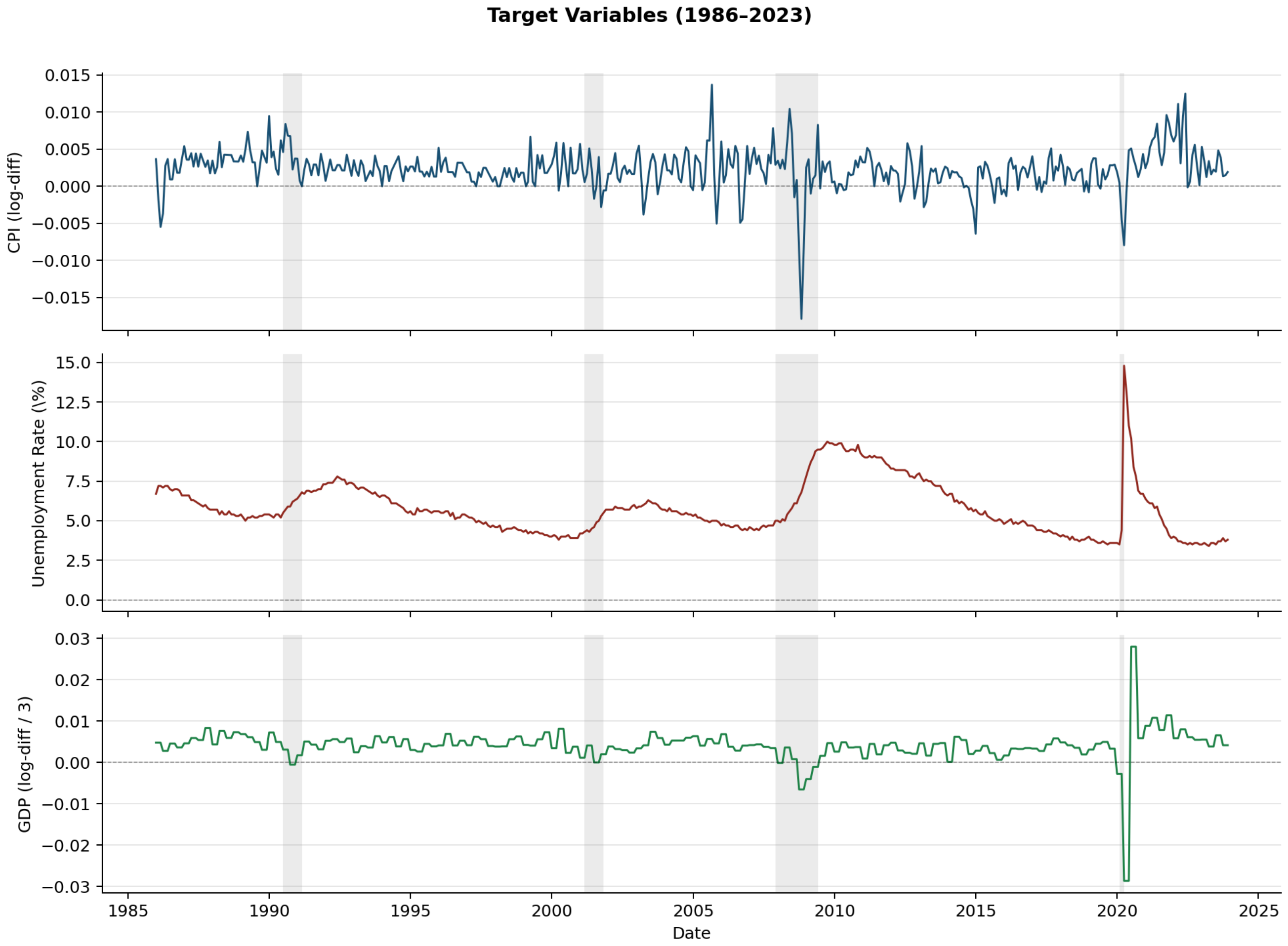
1. **Target variables**
   - CPI, UNRATE, GDP (quarterly: forward-fill missing months) $\Rightarrow$ AR and ARIMA only take targets
2. **Covariates**
   - Monthly variables: payroll (PAYEMS), industrial production (INDPRO), hours worked in manufacturing (AWHMAN), OECD composite leading indicator (USACLI)
   - Weekly / daily variables: exchange rates (GBP, YEN), effective federal funds rate (FFR), Fed balance sheet (WALCL)
     - Higher-frequency variables: take mean **and** std per month (volatility)
   - Lags at 1, 2, 6, 12 months for all target variables
3. **Preprocessing**: log-difference of CPI, GDP, PAYEMS, INDPRO, AWHMAN

Inspect Data Frame:

date	CPI	PAYEMS	INDPRO	UNRATE	AWHMAN	USACLI	GDP	GBP_mean	GBP_std	YEN_mean	YEN_std	FFR_mean
1986-01	0.0036	0.0012	0.0055	6.7	-0.0049	0.0002	0.0048	1.4244	0.0227	199.8905	3.6638	8.1448
1986-02	-0.0018	0.0012	-0.007	7.2	0.0	0.0001	0.0048	1.4297	0.0352	184.8516	4.586	7.86
1986-03	-0.0055	0.0009	-0.0071	7.2	0.0	-0.0001	0.0048	1.4674	0.0154	178.6938	1.8252	7.479
1986-04	-0.0037	0.0019	0.0014	7.1	-0.0049	-0.0001	0.0028	1.4985	0.0363	175.0918	5.1994	6.988
1986-05	0.0028	0.0013	0.0015	7.2	0.0049	-0.0002	0.0028	1.5211	0.021	167.0314	3.4488	6.8529
1986-06	0.0037	-0.0009	-0.004	7.2	0.0	-0.0003	0.0028	1.5085	0.0157	167.5419	2.5161	6.9167

FFR_std	T10Y2Y_mean	T10Y2Y_std	dissent_rate_d	dissent_net_hawk	dissent_net_hawk_s	n_tighter_sum	n_easier_sum	any_dissent_recen	days_since_fomc	days_to_fomc	fomc_cycle_pos	meeting_this_mon
1.0103	1.0514	0.0368	0.0922	0.25	2.0	5.0	3.0	1.0	15.0	42.0	0.2632	0.0
0.1221	0.7368	0.2171	0.0922	0.25	2.0	5.0	3.0	1.0	46.0	11.0	0.807	1.0
0.2745	0.564	0.0644	0.1131	0.0	0.0	5.0	5.0	1.0	17.0	31.0	0.3542	0.0
0.2155	0.6041	0.0734	0.1292	0.0	0.0	5.0	5.0	1.0	0.0	49.0	0.0	1.0
0.085	0.6424	0.0375	0.1131	0.0	0.0	5.0	5.0	1.0	30.0	19.0	0.6122	1.0
0.175	0.6148	0.1007	0.114	0.0	0.0	5.0	5.0	1.0	12.0	38.0	0.24	0.0

Inspect Target Vars

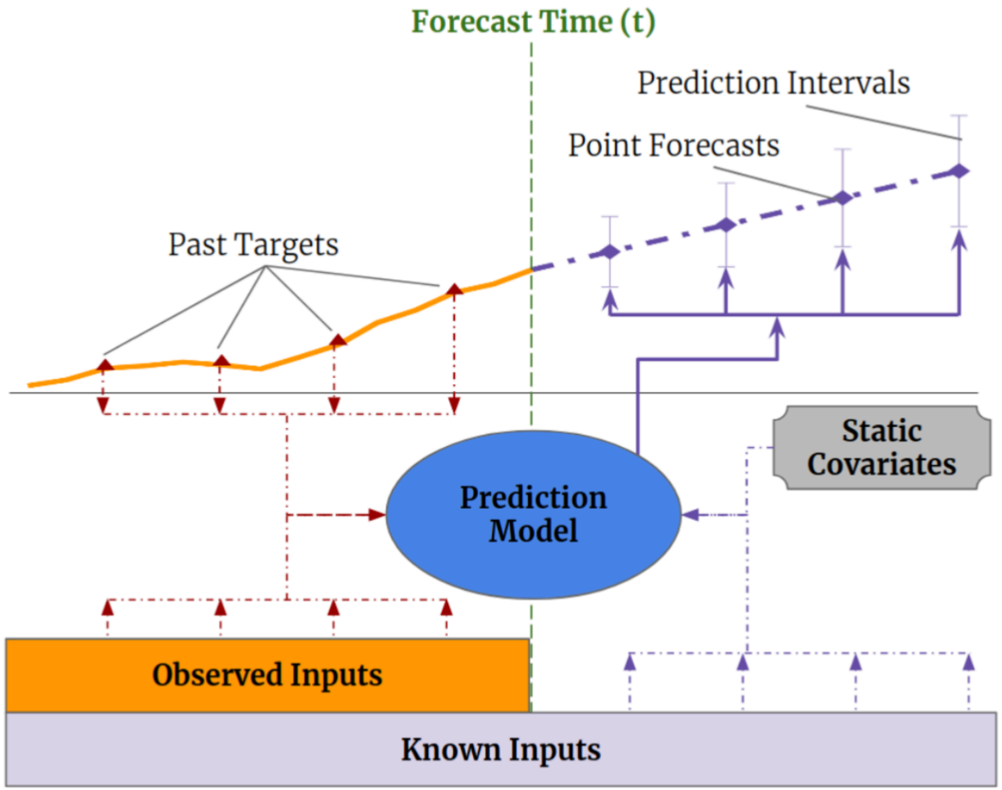


Variable List

The TFT takes four different types of variables:

1. **Static categoricals**: time-invariant, categorical
   - Identifier and frequency of each time series
2. **Static reals**: time-invariant, continuous
   - Number downloads time series from FRED, how long time series has existed
3. **Time-Varying Known Reals**: time-varying, known
   - Day of week, month, year; if day is holiday
   - FOMC Meeting dates
4. **Time-Varying Unknown Reals**: time-varying, unknown
   - Macroeconomic variables, speech embeddings, FOMC dissents

TFT: Input Types



Dimensionality Reduction

FinBERT and FOMC-RoBERTa embeddings are of dimension 768 per speech

$\Rightarrow$ Dimensionality reduction needed for TFT input

We test two methods (number of components X is hyperparameter, fit only on training data):

1. **PCA**: projects embeddings onto X principal components
   - 20 components $\approx$ 80% of variance explained
2. **Factor Analysis (FA)**: models embeddings as linear combination of X latent factors
   - Unlike PCA: explicitly models noise, focuses on **shared** variance
   - Parameters estimated via EM algorithm

Based on your tuning code, here are the two appendix slides filled out: typst

Hyperparameter Tuning: Stage 1 Search Space

Architecture tuned on macro-only TFT (per target × horizon):

Parameter	**Type**	**Range / Values**
Encoder length	int	12 – 48
Hidden size	categorical	$\{8, 16, 32, 64, 128, 256\}$
Hidden continuous size	categorical	$\{2, 4, 8, 16\}$
Dropout	float	0.05 – 0.55
Learning rate	log-uniform	10^{-4} – 0.15
Normalizer	categorical	{encoder-none, group}

Fixed: `lstm_layers = 2`, `attention_head_size = 2`, `max_epochs = 50`, `patience = 15`

Hyperparameter Tuning: Stage 2 Search Space

Speech embedding params tuned (architecture fixed from Stage 1):

Parameter	Type	Range / Values
Aggregation	categorical	{mean, decay, attention}
Reduction	categorical	{pca, fa}
PCA components (X)	int	5 – 30
Speech window (months)	int	3 – 12

Tuned separately for each of 9 combinations: {CPI, GDP, UNRATE} $\times$ {3, 6, 12} months

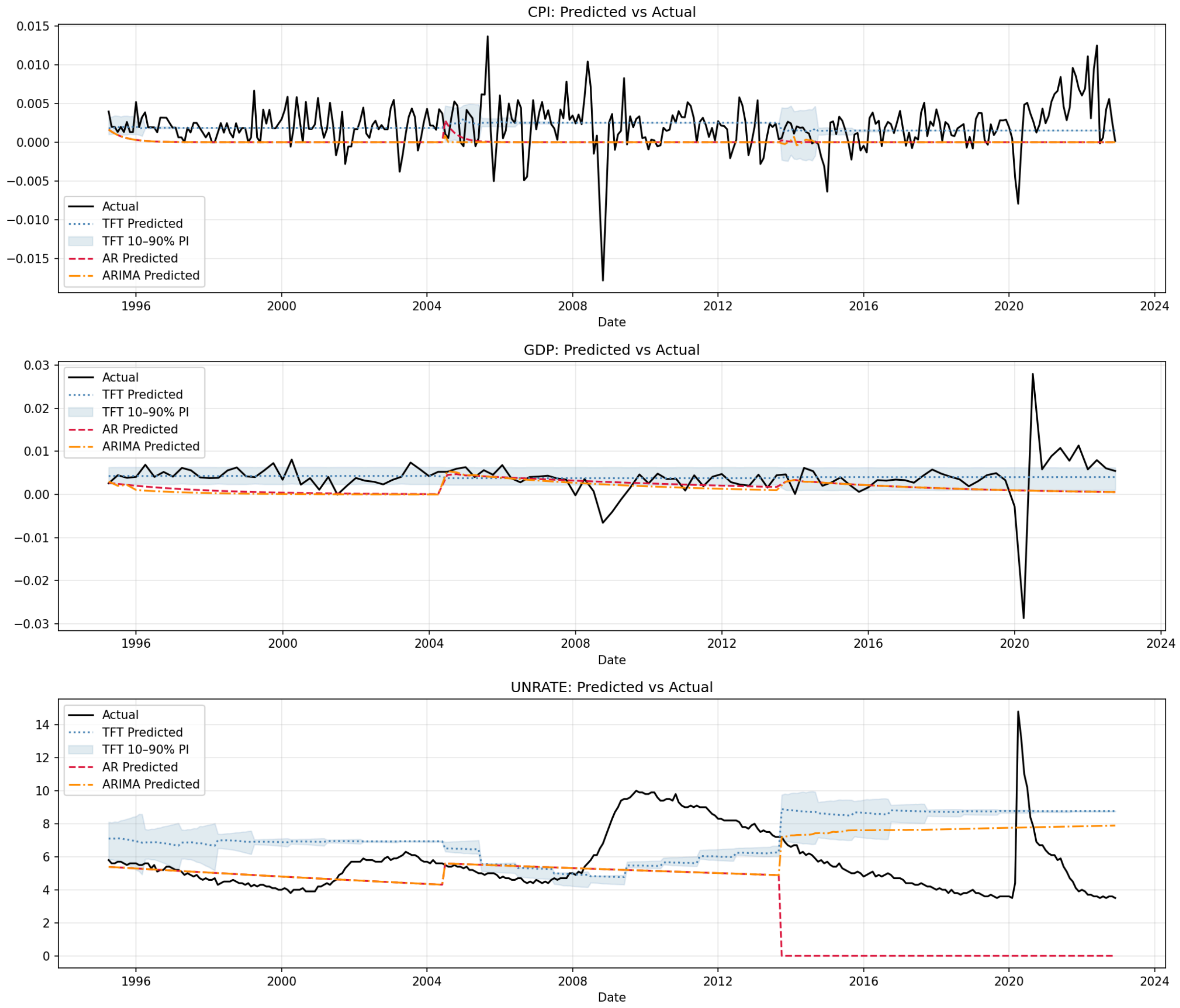
Hyperparameter Tuning: Stage 2 – Optimal Hyperparameters

	FOMC-RoBERTa			**FinBERT**		
Target	**h=3**	**h=6**	**h=12**	**h=3**	**h=6**	**h=12**
CPI	Mean PCA, n = 21 W = 3 MAE = 0.001847	Attention FA, n = 10 W = 8 MAE = 0.001849	**Attention** **PCA, n = 12** **W = 9** **MAE = 0.001842**	**Attention** **FA, n = 20** **W = 5** **MAE = 0.001846**	**Decay** **PCA, n = 23** **W = 3** **MAE = 0.001846**	Mean PCA, n = 8 W = 7 MAE = 0.001847
GDP	**Mean** **PCA, n = 8** **W = 7** **MAE = 0.00154**	Attention FA, n = 12 W = 8 MAE = 0.00171	Attention FA, n = 5 W = 10 MAE = 0.00152	Attention FA, n = 6 W = 12 MAE = 0.00170	**Attention** **FA, n = 10** **W = 8** **MAE = 0.00157**	**Decay** **PCA, n = 20** **W = 12** **MAE = 0.00151**
UNRATE	Decay FA, n = 12 W = 11 MAE = 1.31293	Attention FA, n = 6 W = 12 MAE = 1.31501	**Mean** **PCA, n = 8** **W = 7** **MAE = 1.11597**	**Mean** **FA, n = 23** **W = 5** **MAE = 1.26152**	**Mean** **PCA, n = 14** **W = 4** **MAE = 1.30896**	Mean PCA, n = 25 W = 4 MAE = 1.20808

where W refers to the speech window size and n to the optimal number of components for the dimensionality reduction. The best result per target$\times$horizon is shown in bold.

Results: In Sample

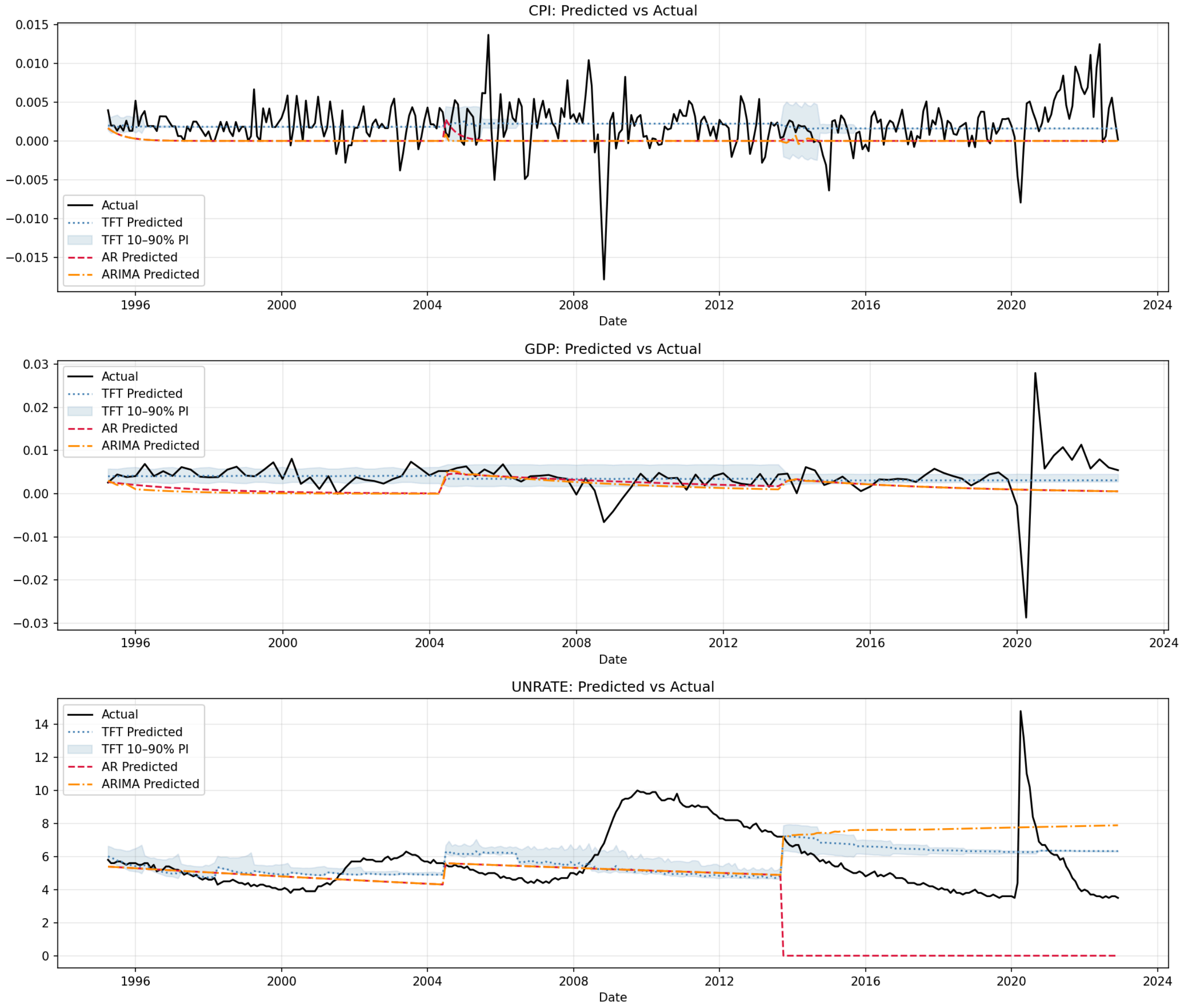
Macro Only: h=12



Macro Only: h=12

Model	Target	MAE ↓	RMSE ↓
AR	CPI	0.00268	0.00348
ARIMA	CPI	0.00268	0.00349
TFT	**CPI**	**0.00196**	**0.00286**
AR	GDP	0.00331	0.00475
ARIMA	GDP	0.0035	0.00489
TFT	**GDP**	**0.00218**	**0.00389**
AR	UNRATE	2.661	2.99886
ARIMA	**UNRATE**	**1.93349**	**2.25383**
TFT	UNRATE	2.56896	2.81994

with Embeddings: h=12



with Embeddings: h=12

Model	Target	MAE ↓	RMSE ↓
AR	CPI	0.00268	0.00348
ARIMA	CPI	0.00268	0.00349
TFT	**CPI**	**0.00195**	**0.00284**
AR	GDP	0.00331	0.00475
ARIMA	GDP	0.0035	0.00489
TFT	**GDP**	**0.00223**	**0.00389**
AR	UNRATE	2.661	2.99886
ARIMA	UNRATE	1.93349	2.25383
TFT	**UNRATE**	**1.66499**	**1.94898**